

Stats 500, Fall 2025

- Instructor:** Gongjun Xu, gongjun@umich.edu
Office hour and Location: MW 11:30am-12:30pm, 462 West Hall
- GSI:** Peiyao Cai, pycai@umich.edu
Office hours and Location:
• TBA
- Web Page:** Canvas
- Text:** Linear Models with R (second edition), by Julian Faraway.
Extending the Linear Model with R, Julian Faraway (Online)
(Online available through the UM library)
- Course Description:** This class will focus on linear and generalized linear models, and will include topics such as building regression models and conducting inference and model checking, variable selection, shrinkage estimators (e.g. ridge regression, LASSO), generalized linear models, etc. This course aims to introduce the basic concepts and statistical methods that have been commonly used in practice. After finishing this course, students should have the basic skills to analyze data sets collected under simple and standard designs.
- Prerequisites:** Matrix algebra, introductory probability and statistics. If in doubt, talk to the instructor. Some familiarity with programming (especially R) is helpful, but not required.
- Grading:** **Option 1** Homework (30%), midterms (40%), individual final project (30%)
Option 2 Homework (30%), midterms (30%), individual final project (40%)
We will take the maximum of the two options.
- Exam:** Midterm Exam: In-class exam, Wednesday, October 8

For exam, you may take one (8.5 by 11 inches; both sides) paper as a cheat sheet.

Excused absences from completing the exams will be handled by the instructor. However, to make up for the absence, you have to (1) inform the instructor about your absence before the day of the exam unless there is an unexpected accident (2) have a legitimate excuse according to the university regulation.
- Hw and Project:** **Homework** is due online. No late homework will be accepted (or accepted with penalty if approved by GSI/Instructor).

Final project details will be announced after the midterm.

Academic Misconduct

Academic dishonesty of any form will not be tolerated. The University of Michigan community functions best when its members treat one another with honesty, fairness, respect, and trust. The college promotes the assumption of personal responsibility and integrity, and prohibits all forms of academic dishonesty and misconduct. All cases of academic misconduct will be referred to the LSA Office of the Assistant Dean for Undergraduate Education. Being found responsible for academic misconduct will usually result in a grade sanction, in addition to any sanction from the college. For more information, including examples of behaviors that are considered academic misconduct and potential sanctions, please see lsa.umich.edu/lsa/academics/academic-integrity.html

Accommodations for Students with Disabilities

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact Services for Students with Disabilities (<https://ssd.umich.edu/>; 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please contact your SSD Coordinator or visit SSD's Current Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience.

Mental Health and Well-Being

University Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressures and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc. If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers a variety of resources, many which are listed on the Resources for Student Well-being webpage. You can also search for additional well-being resources on that website.